Artifact 4

Navigating Data Challenges in Machine Learning

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Navigating Data Challenges in Machine Learning

How real-world data problems - messy records, imbalance, privacy, and drift - decide whether a model is trustworthy, and how I'd handle each one in production.

Objective: Adapted from the Workshop 4 Data Challenge Scenarios activity, this artifact demonstrates the ability to identify and address the practical data problems that decide whether a machine learning model is accurate, fair, and reliable in production. It covers data cleaning, class imbalance, model selection and justification, false-positive vs false-negative trade-offs, data privacy, bias mitigation, and post-deployment monitoring. This competency is distinct from Artifacts 1-3: not building an AI experience, selecting a model, or explaining a concept - but operationalizing ML responsibly, the skill that determines whether a deployed model holds up under real data.

Data Cleaning & Preprocessing

Missing Values, Outliers, and Imbalance

Real ML data is messy. Missing values are dropped only when rare and random; otherwise I impute (mean/median) or flag them so the model knows they were absent. Outliers are investigated before deletion - a $9,999 transaction may be fraud, not an error. Class imbalance is the quiet killer: if 98% of tickets are "resolved," a model that always predicts "resolved" scores 98% accuracy yet catches nothing. I handle it with class weights, resampling (SMOTE), and metrics beyond accuracy (precision, recall, F1).

Automation example: in bot-failure prediction, "failure" events are rare against a sea of "success" logs. Naive accuracy hides the model's real weakness, so I weight the failure class and report recall, not just accuracy.

Model Selection & Evaluation

Model choice follows the problem: structured, tabular, explainability-sensitive data -> gradient boosting or logistic regression; raw images/audio -> deep learning. I justify by data type, volume, latency, and whether auditors must see why . A less-accurate-but-explainable model is often the right call in regulated banking.

Accuracy lies on imbalanced data. A false positive is a false alarm; a false negative is a missed event. Which hurts more is domain-specific. In RPA exception handling, a false "auto-resolved" (FN) can post a bad entry, so I tune toward fewer FNs even if false alarms (FP) rise - because a human can absorb extra review, but not an undetected error.

Data Privacy, Security & Bias

Training data often carries PII - customer names, account context. Risks include re-identification, leakage, and regulatory exposure (GDPR/CCPA, banking controls). I minimize columns, tokenize/pseudonymize, restrict access, and use synthetic data for development, keeping training data inside the regulated environment.

Biased historical data yields a biased model. I audit training data for representation and check outcomes across groups. If past "success" labels come only from one business unit, the model learns that unit's pattern as "normal" and misreads others - so I verify coverage before training, not after.

Post-Deployment Monitoring

Models Drift - Verification Never Ends

A model degrades as data changes. My monitoring plan tracks input distributions, accuracy, and false-positive/false-negative rates over time; sets alert thresholds; schedules retraining; keeps a human review on high-stakes outputs; and logs overrides - where humans correct the model. Overrides are where hidden bias and drift surface first.

Automation example: I already monitor 52 bots in production for failure drift. A model needs the same observability - dashboards on prediction volume, score distribution, and override rate. The lifecycle doesn't end at deployment; that's where governance begins.

Process: Working With the AI Coach

Skills Demonstrated

Artifacts 1-3 show I can design, select, and explain AI. Artifact 4 shows I can operate it responsibly - cleaning messy data, choosing the right metric, protecting privacy, and monitoring after launch. In enterprise automation, that operational discipline is what separates a demo from a system people trust. Accuracy doesn't earn trust; reliability under real data does.

This version is tailored for ML engineering leads, risk/compliance partners, and hiring managers evaluating production-readiness - not just model trivia. The original activity was a guided chat; I revised it into a structured competency page with banking/automation examples, showing I can turn a learning exercise into an operational standard. The framing reflects my COE experience: govern the model like you govern a production bot.